1 [30pt] System

Consider the unicycle 4-state model. The state $x = [p_x, p_y, v, \theta]^T \in \mathbb{R}^4$ contains the 2D position (p_x, p_y), the speed v, and the heading θ of the vehicle. The control input $u = [\dot{v}, \dot{\theta}]^T \in \mathbb{R}^2$ contains the acceleration $\dot{v}$ and the turning rate $\dot{\theta}$. The dynamic equation is

$$\dot{x} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} u \tag{1}$$

1.1 [1pt] Is this system control-affine?

Solution: Yes, the system is control affine because the control input shows up linearly in the dynamics.

1.2 [5pt] Which description is most appropriate? Choose one from each row

Linear	Nonlinear
Continuous-Time	Discrete-Time
Time-invariant	Time-varying
Deterministic	Stochastic
Continuous-State	Discrete-State

Solution: The following description is appropriate:

- Nonlinear
- Continuous-Time
- Time-invariant
- Deterministic
- Continuous-State

1.3 [8pt] Linearize the system at $[p_x^r, p_y^r, v^r, \theta^r]^T$. Write down the resulting dynamics.

Solution: The resulting dynamics are defined by

$$\delta\dot{x} = \begin{bmatrix} 0 & 0 & \cos\theta^r & -v\sin\theta^r \\ 0 & 0 & \sin\theta^r & v\cos\theta^r \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \delta x + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \delta u$$

where $\delta x = x - x^r$, $x^r = [p_x^r, p_y^r, v^r, \theta^r]^T$, and $\delta u = u - u^r$.

1.4 [8pt] Compute the linearized system dynamics at $[1, 1, 0, 0]^T$. Is the linearized system at this point controllable? Justify your answer.

Solution: The linearized dynamics at $[1, 1, 0, 0]^T$ are

$$\begin{aligned}
\delta\dot{x} &= \begin{bmatrix} 0 & 0 & \cos\theta^r & -v\sin\theta^r \\ 0 & 0 & \sin\theta^r & v\cos\theta^r \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \delta x + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \delta u && (2) \\
&= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \delta x + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \delta u. && (3)
\end{aligned}$$

This linearized system is not controllable because nothing affects the p_y state. This will show up as a deficient rank of the controllability matrix.

1.5 [8pt] Under which condition on the parameters $p_x^r, p_y^r, v^r, \theta^r$ is the linearized system controllable? Find sufficient and necessary conditions.

Solution: The linearized system is controllable if and only if $\text{rank}(\mathcal{C}) = n$, where $\mathcal{C}$ is the controllability matrix and n is the number of the dimensions. Since $n = 4$ for the unicycle model, we know the controllability matrix is

$$\mathcal{C} = \begin{bmatrix} A^3B & A^2B & AB & B \end{bmatrix}.$$

Evaluating this expression, we arrive at

$$\mathcal{C} = \left[\begin{array}{cc|cc|cc|cc} 0 & 0 & 0 & 0 & \cos\theta^r & -v^r\sin\theta^r & 0 & 0 \\ 0 & 0 & 0 & 0 & \sin\theta^r & v^r\cos\theta^r & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Since the first 4 columns are zero, we can investigate the "right" 4×4 square matrix. From the 2 bottom rows, we see they are immediately linearly independent, giving an immediate $\text{rank}(\mathcal{C}) \geq 2$. Thus, we can further simplify the problem and investigate the rank of

$$\mathcal{M} = \begin{bmatrix} \cos\theta^r & -v^r\sin\theta^r \\ \sin\theta^r & v^r\cos\theta^r \end{bmatrix}$$

If the determinant of $\mathcal{M}$ is non-zero, then $\mathcal{M}$ itself is full rank, and $\text{rank}(\mathcal{C}) = 4$. Thus, we evaluate

$$\begin{aligned}
\det(\mathcal{M}) &= v^r\cos^2\theta^r + v^r\sin^2\theta^r \\
&= v^r(\cos^2\theta^r + \sin^2\theta^r) \\
&= v^r
\end{aligned}$$

Thus, $\det(\mathcal{M}) \neq 0$ if and only if $v^r \neq 0$, and therefore, the system is controllable if and only if $v^r \neq 0$.

2 [46pt] Model Mismatch

In this section, the controller imagines that the vehicle follows the unicycle 4-state model

$$x_u = \begin{bmatrix} p_x \\ p_y \\ v \\ \theta \end{bmatrix}, \qquad u_u = \begin{bmatrix} \dot{v} \\ \dot{\theta} \end{bmatrix}, \tag{4}$$

where (p_x, p_y) is the position in the fixed world frame, v is the forward speed, θ is the heading, $\dot{v}$ is the longitudinal acceleration, and $\dot{\theta}$ is the turning rate. The actual rollout uses the dynamic bicycle model supplied in `hw1/vehicle_models.py`. Both transitions use forward Euler integration with time step $\Delta t = 0.01\,\mathrm{s}$. Unless otherwise stated, use

$$x_u(0) = \begin{bmatrix} 0 \\ 0 \\ 0.5 \\ 0 \end{bmatrix}, \qquad p_g = \begin{bmatrix} 5 \\ 5 \end{bmatrix}, \qquad T = 10\,\mathrm{s}. \tag{5}$$

Here, $x_u(0)$ is the initial unicycle state, p_g is the goal position with components $p_{g,x}$ and $p_{g,y}$, and T is the rollout duration.

2.1 [15pt] Unicycle feedback controller. Define the goal-position error $e_p \in \mathbb{R}^2$, distance to the goal $\rho \in \mathbb{R}_{\geq 0}$, desired heading θ_d, and wrapped heading error e_θ by

$$e_p = \begin{bmatrix} p_{g,x} - p_x \\ p_{g,y} - p_y \end{bmatrix}, \qquad \rho = \|e_p\|_2, \tag{6}$$

$$\theta_d = \tan^{-1}\left(\frac{e_{p,y}}{e_{p,x}}\right), \qquad e_\theta = \mathrm{wrap}(\theta_d - \theta) \in [-\pi, \pi). \tag{7}$$

Here, θ_d is the angle of the position-error vector measured from the positive x-axis. The operator $\mathrm{wrap}(\phi)$ maps an angle ϕ to $[-\pi, \pi)$. In code, compute this operation with the provided `wrap_angle()` function; do not implement a separate angle-wrapping rule.

Let $k_v > 0$ be the velocity-feedback gain, $k_\theta > 0$ be the heading-feedback gain, $v_{\max} > 0$ be the maximum desired speed, and v_d be the desired speed defined by

$$v_d = \min(v_{\max}, \rho)\max(0, \cos e_\theta), \tag{8}$$

Implement the two controller calculations

$$u_u = \begin{bmatrix} \dot{v} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} k_v(v_d - v) \\ k_\theta e_\theta \end{bmatrix}. \tag{9}$$

Complete `unicycle_control` in `problem.py`. Use the default gains $(k_v, k_\theta) = (2.0, 2.5)$, run the unicycle model, and submit the XY trajectory and every state-versus-time plot. Explain the purpose of each gain and of the factor $\max(0, \cos e_\theta)$. The supplied return lines already clip the acceleration and turning rate. After computing the errors and desired speed, you only need to calculate `v_dot` and `theta_dot`; do not add another saturation operation.

Solution: Please see the generated plots in Figure 1 and Figure 2. The purpose of the gains and the factor $\max(0, \cos e_\theta)$ are detailed in the following bullet points.

- k_v: This gain provides a feedback multiplication on the velocity error. Thus, this gain drives the response in velocity tracking.
- k_θ: This gain provides a feedback multiplication on the heading error. Thus, this gain drives the response in heading tracking.
- $\max(0, \cos e_\theta)$: This maximum protects against commanding a positive speed when the vehicle is pointed more than $\pi/2$ radians away from the goal. In this case, we'd rather only run our heading towards to target to wait until we are headed in the correct direction. Otherwise, we are not moving *towards* the goal position.

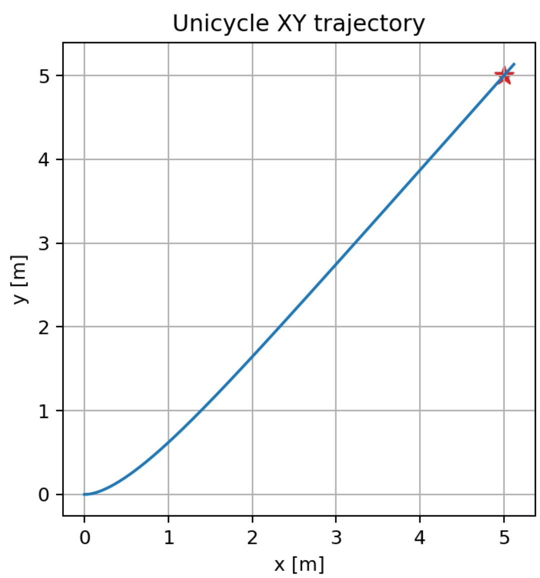

Figure 1: XY trajectory plot.

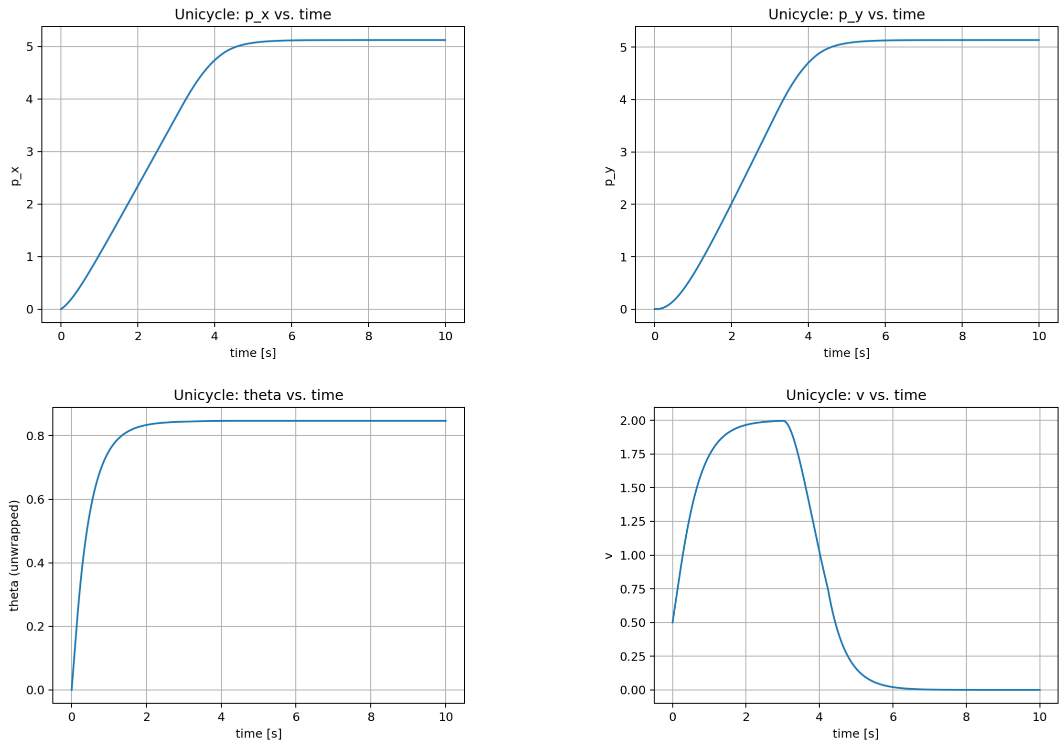

Figure 2: State-versus-time plots.

2.2 [15pt] Pure-rolling bicycle command conversion. The dynamic bicycle state and control follow the Lecture 2 and SPARK order

$$x_b = \begin{bmatrix} X \\ Y \\ v_x \\ v_y \\ r \\ \psi \end{bmatrix}, \qquad u_b = \begin{bmatrix} a_x \\ \delta \end{bmatrix}, \tag{10}$$

where (X, Y) is the position in the fixed world frame, v_x and v_y are the longitudinal and lateral body-frame velocities, $r = \dot{\psi}$ is the yaw rate, ψ is the heading, a_x is the longitudinal acceleration command, and δ is the front-wheel steering angle. Implement the initial-state conversion

$$x_b(0) = \begin{bmatrix} p_x(0) \\ p_y(0) \\ v(0) \\ 0 \\ 0 \\ \theta(0) \end{bmatrix}. \tag{11}$$

At each step, evaluate the Problem 2.1 controller using the virtual unicycle state $\widetilde{x}_u$, defined by

$$\widetilde{x}_u = \begin{bmatrix} X \\ Y \\ v_x \\ \psi \end{bmatrix}. \tag{12}$$

The actual rollout plant remains the six-state SPARK dynamic bicycle, whose lateral motion is generated by front and rear tire forces. For the purpose of converting the unicycle command only, replace those tire-force equations by the no-side-slip, pure-rolling approximation. Here, α_f and α_r denote the front and rear tire slip angles, and $l_f > 0$ and $l_r > 0$ denote the distances from the vehicle center of mass to the front and rear axles, respectively. The pure-rolling constraints are

$$\alpha_f = \alpha_r = 0 \quad \Longrightarrow \quad v_y - l_r r = 0, \qquad v_y + l_f r = v_x \tan\delta. \tag{13}$$

With wheelbase $L = l_f + l_r$, these constraints give

$$r = \dot{\psi} = \frac{v_x}{L}\tan\delta. \tag{14}$$

Therefore, if the shared controller returns $(\dot{v}, \dot{\theta})$, implement the provided command conversion

$$u_b = \begin{bmatrix} a_x \\ \delta \end{bmatrix} = \begin{bmatrix} \dot{v} \\ \operatorname{sat}_{\delta_{\max}}\left(\tan^{-1}\left(\frac{L\dot{\theta}}{v_x}\right)\right) \end{bmatrix}. \tag{15}$$

Here, $\delta_{\max} > 0$ is the steering-angle limit and $\operatorname{sat}_{\delta_{\max}}(z) = \min(\delta_{\max}, \max(-\delta_{\max}, z))$.

The inverse tangent is taken on its principal branch, and the conversion is used on the forward-motion domain $v_x > 0$. The acceleration command already satisfies its limit from Problem 2.1. Use the pure-rolling relation only for the command conversion: do not replace the supplied tire-force bicycle transition used for the actual rollout.

Complete `bicycle_initial_state` and `bicycle_control`, run the bicycle model, and submit the XY trajectory and every state-versus-time plot. Explain the tracking error caused by model mismatch. Hint: relate the error to the bicycle's lateral velocity v_y and yaw rate r.

Solution: Please see the generated plots in Figure 3 and Figure 5. The main gap in this model is the no-slip, pure-rolling approximation. If this model were to hold, we would see $v_y = l_r r$ for the entirety of the trajectory. This would mean there is not slip occurring in the true, force-based model. Plotting to see if $v_y = l_r r$ holds in Figure 4, we can see that at the beginning of the trajectory (where the heading error is large), this no-slip condition is in fact violated. This induces tracking error because the true heading is different from what the controller intends, causing drift in position.

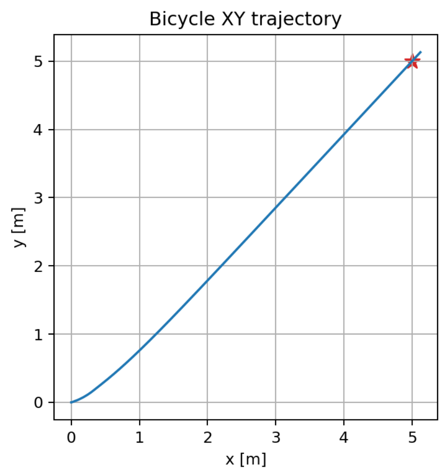

Figure 3: XY trajectory plot.

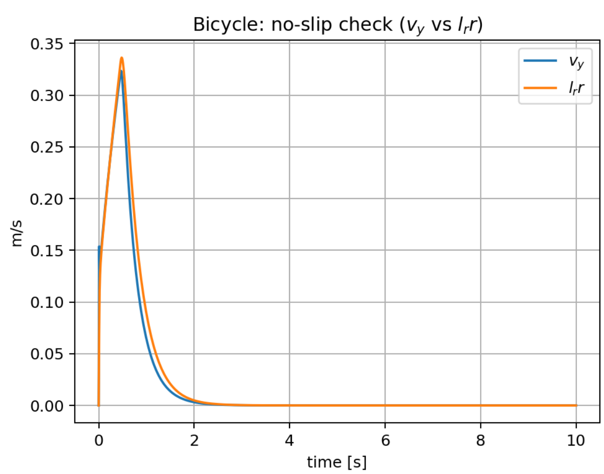

Figure 4: No-slip assumption plot.

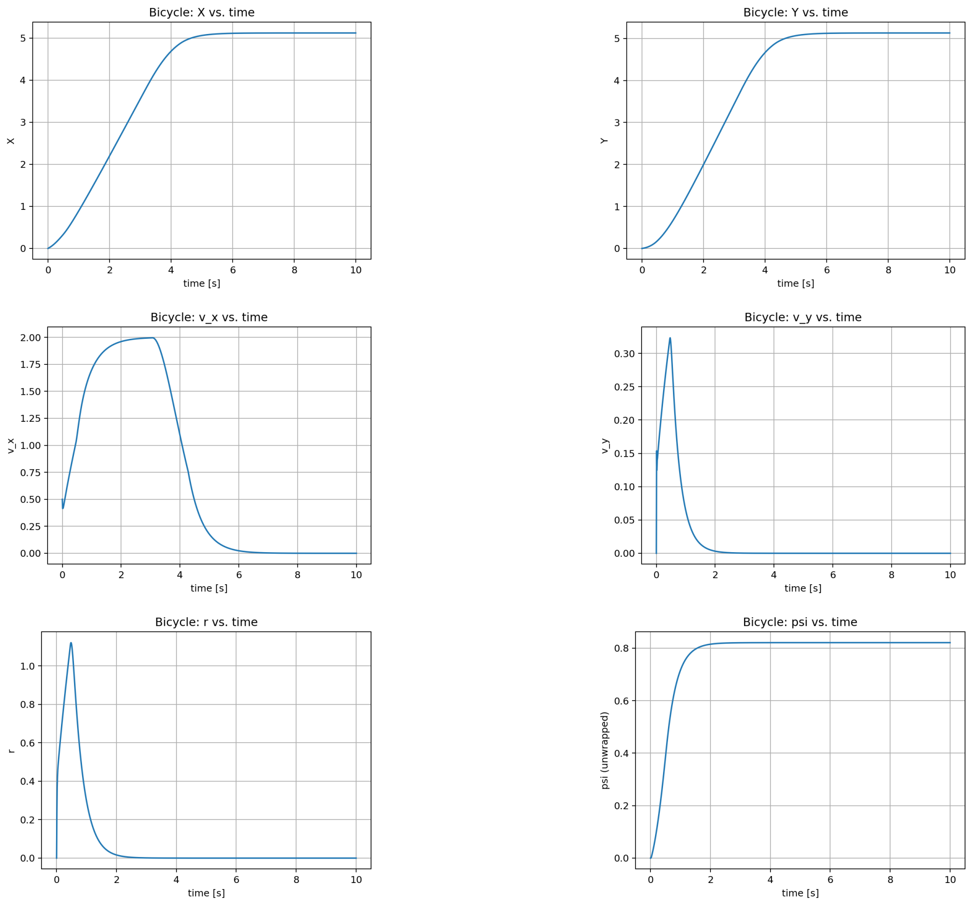

Figure 5: State-versus-time plots.

2.3 [16pt] Two-parameter performance analysis. Use the shared controller for both models and evaluate all nine gain pairs in

$$k_v \in \{1.5, 2.0, 2.5\}, \qquad k_\theta \in \{1.5, 2.5, 3.5\}. \tag{16}$$

At matching sample indices k, corresponding to time $t_k = k\Delta t$, define the unicycle–bicycle position mismatch $e_p^{\mathrm{mm}}(k)$ and orientation mismatch $e_\theta^{\mathrm{mm}}(k)$ by

$$e_p^{\mathrm{mm}}(k) = \left\| \begin{bmatrix} p_{x,k} \\ p_{y,k} \end{bmatrix} - \begin{bmatrix} X_k \\ Y_k \end{bmatrix} \right\|_2, \qquad e_\theta^{\mathrm{mm}}(k) = \left| \mathrm{wrap}(\theta_k - \psi_k) \right|. \tag{17}$$

Here, $(p_{x,k}, p_{y,k}, \theta_k)$ are components of the unicycle state at t_k, while (X_k, Y_k, ψ_k) are components of the bicycle state at the same time. Use the provided `wrap_angle()` function for the angular difference.

Run the provided gain study and submit exactly two figures: (i) position error versus time and (ii) orientation error versus time. Each figure should show all nine gain pairs with a readable legend. Analyze how k_v and k_θ affect the transient and final errors, and discuss whether one gain pair is uniformly best for both errors.

Solution:
Figures. Please see the generated plots in Figure 6.

Gain discussion. k_v has the larger effect of the two gains on performance, determining whether the system converges without overshooting and having to turn back, where the model difference is much more significant. If the gain is too low, the controller does not keep up with the desired velocity and performs poorly near the final position, not slowing down fast enough. If the gain is high enough, however, there is not substantial difference in performance, only varying slightly in transient characteristics.

k_θ determines how direct the path is the final goal, with higher gains causing more initial overshoot (likely partly due to model mismatch), but having better final error. The lower gains perform better transiently but exhibit more steady-state error at the final orientation.

There is no gain pair that performs the best across all of the errors (for both transient and final), so the choice should be made on desired characteristics. Likely, the most important error is position, so the gain pair $(k_v, k_\theta) = (2.5, 3.5)$ would be a good choice, causing more initial overshoot in orientation but having good overall final error characteristics.

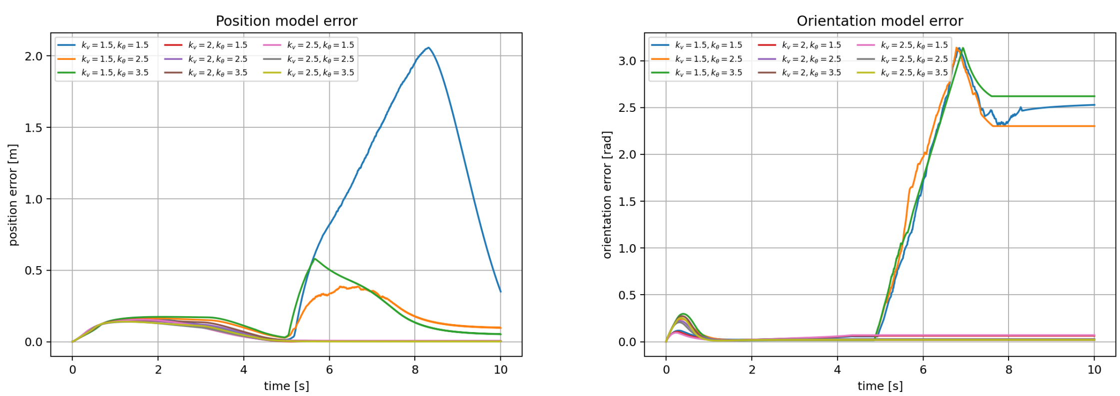

Figure 6: Gain study plots.

3 [24pt] True or False

3.1 [3pt] There is a unique equilibrium point for any system.

Solution: False

3.2 [3pt] For linear systems, local stability implies global stability.

Solution: True

3.3 [3pt] A stable system is bounded.

Solution: True

3.4 [3pt] A stable system converges to its equilibrium.

Solution: False

3.5 [3pt] Adding additional actuators to a system can always improve its controllability.

Solution: False

3.6 [3pt] Unobservability occurs due to system stochasticity.

Solution: False

3.7 [3pt] For a Markov process, the future state depends on the past state even when the current state and control are given.

Solution: False

3.8 [3pt] For any nonlinear system with known model and known initial state, the future can be fully predicted by simulation.

Solution: True